US 20250272186A1

(19) United States
(12) Patent Application Publication
PARK et al.

(10) Pub. No.: US 2025/0272186 A1
(43) Pub. Date: Aug. 28, 2025

(54) STORAGE DEVICE PERFORMING RECOVERY OPERATION BY USING ENDURANCE DATA, AND OPERATION METHOD THEREOF

(71) Applicant: SAMSUNG ELECTRONICS CO., LTD., Suwon-si (KR)

(72) Inventors: SNGHOON PARK, Suwonpsi (KR); JUYEON PARK, Suwon-si (KR); JISUNG BYUN, Suwon-si (KR)

(73) Assignee: SAMSUNG ELECTRONICS CO., LTD., Suwon-si (KR)

(21) Appl. No.: 18/962,959

(22) Filed: Nov. 27, 2024

(30) Foreign Application Priority Data

Feb. 22, 2024 (KR) 10-2024-0025615

Publication Classification

(51) Int. Cl.
G06F 11/07 (2006.01)

(52) U.S. Cl.
CPC *G06F 11/0793* (2013.01); *G06F 11/0727* (2013.01)

(57) ABSTRACT

Provided is an operating method of a storage controller configured to communicate with a non-volatile memory device. The operating method includes receiving a memory response from the non-volatile memory device, generating a first value of a first endurance type and a second value of a second endurance type, based on the memory response, obtaining a third value by applying the first value to a threshold function, the threshold function being defined based on a correlation between the first endurance type and the second endurance type, determining whether the second value is greater than or equal to the third value, and performing a recovery operation in response to determining that the second value is greater than or equal to the third value.

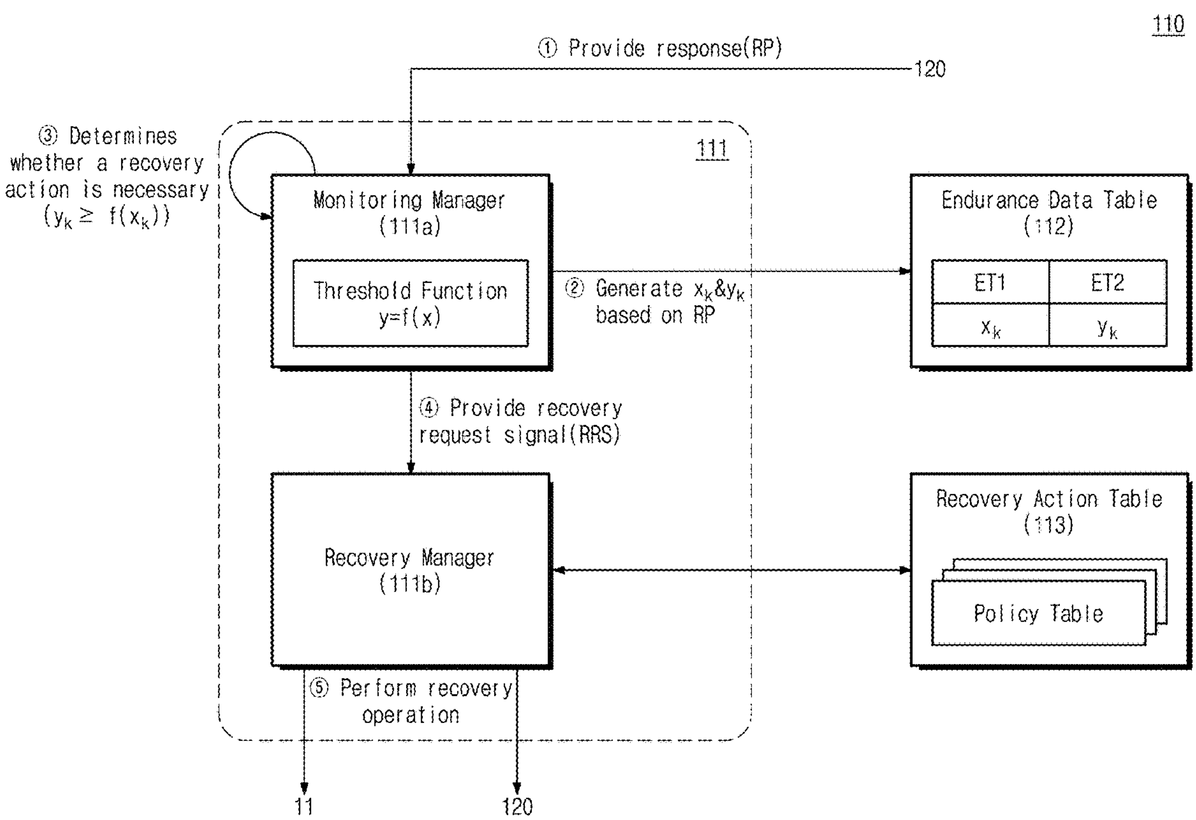


FIG. 1

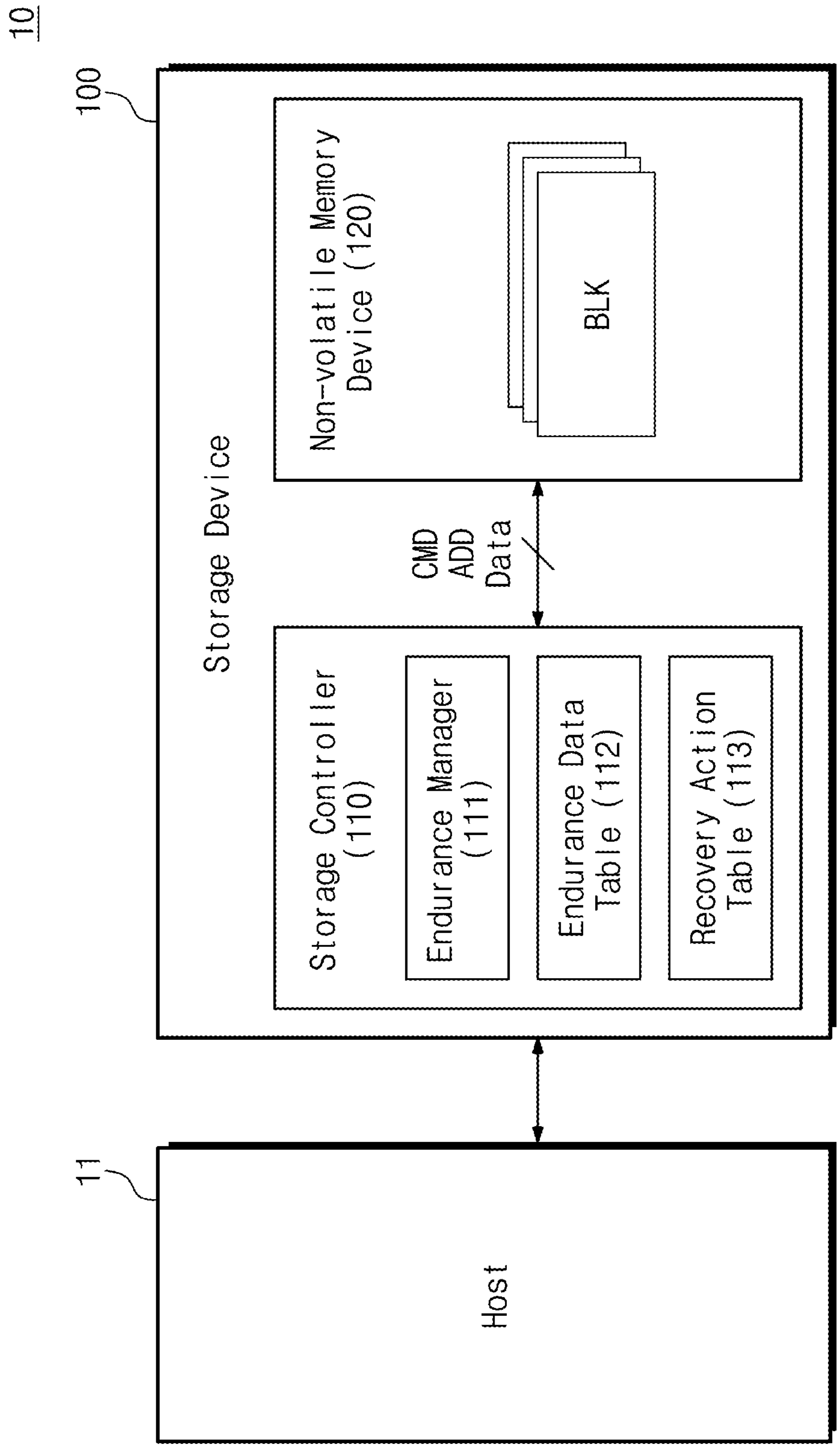

10
100
Storage Device
Storage Controller (110)
Endurance Manager (111)
Endurance Data Table (112)
Recovery Action Table (113)
CMD
ADD
Data
Non-volatile Memory Device (120)
BLK
11
Host

FIG. 2

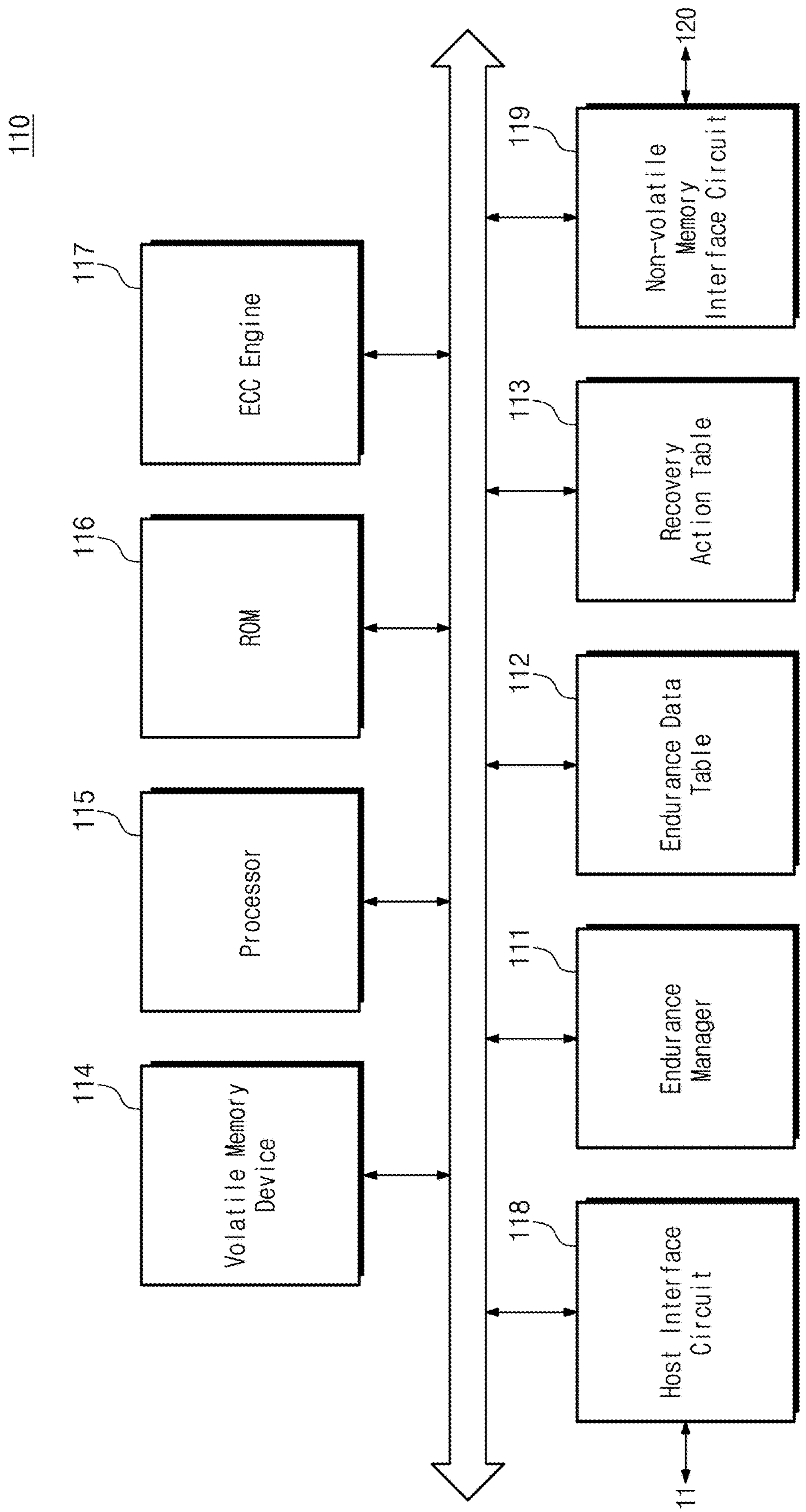

110
114
Volatile Memory Device
115
Processor
116
ROM
117
ECC Engine
118
Host Interface Circuit
111
Endurance Manager
112
Endurance Data Table
113
Recovery Action Table
119
Non-volatile Memory Interface Circuit
120
11

FIG. 3

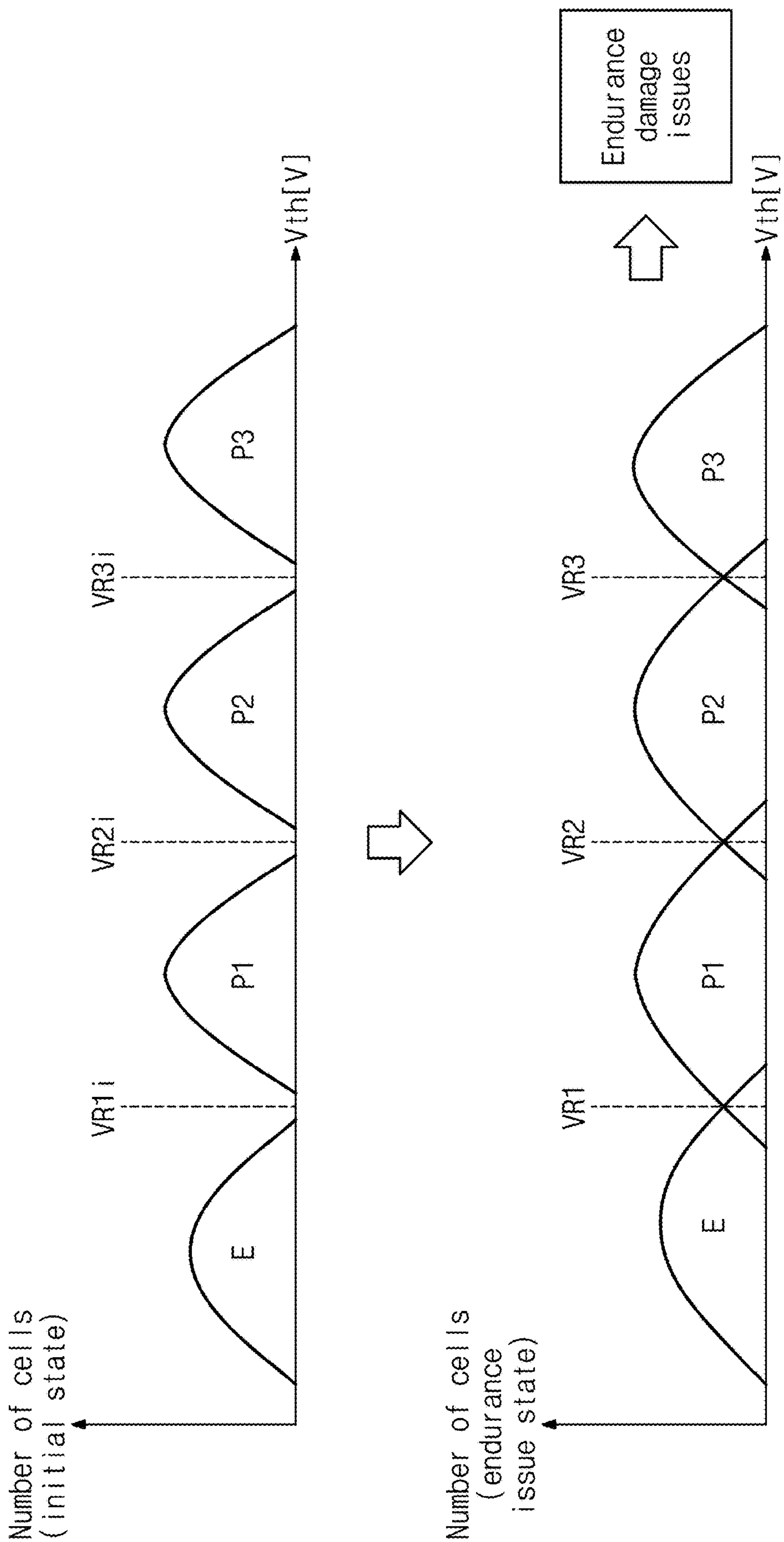
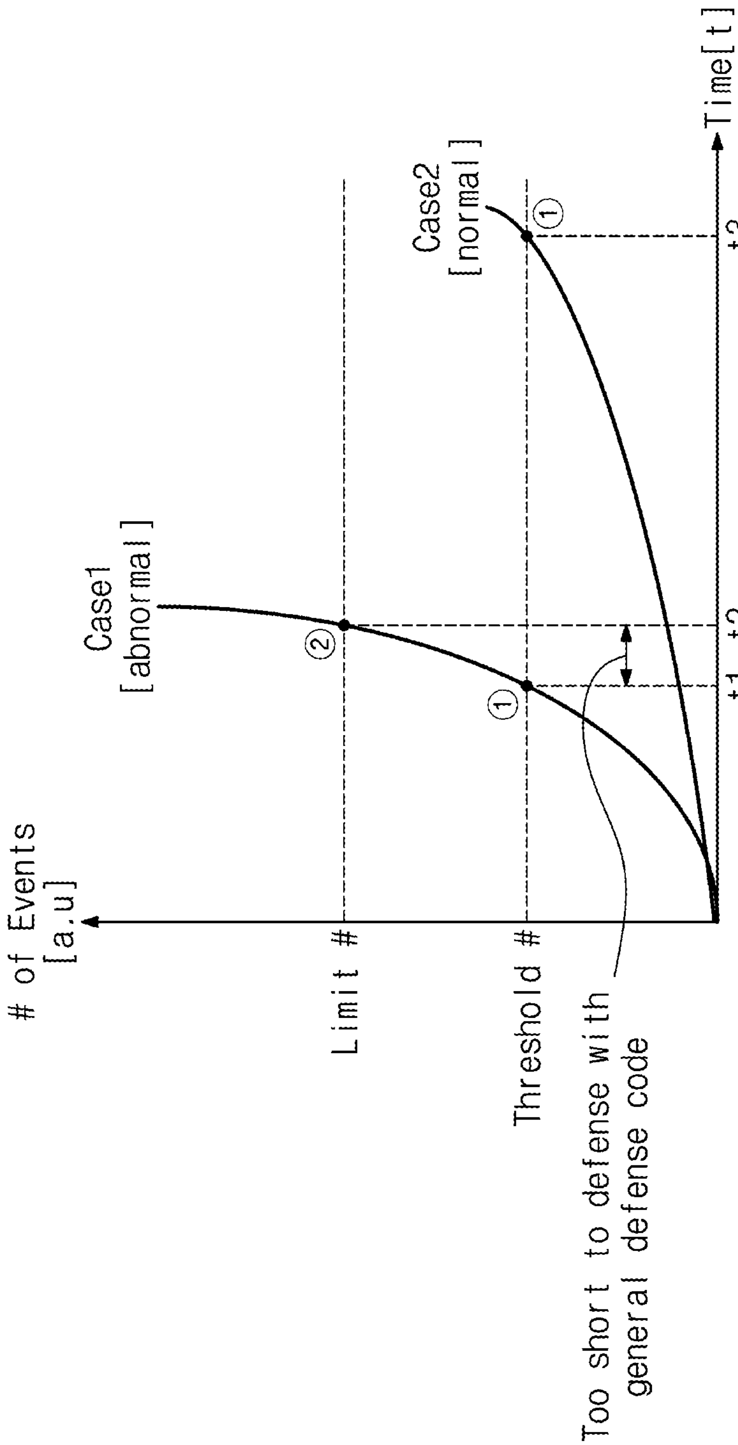

Number of cells
(initial state)
E
VR1i
P1
VR2i
P2
VR3i
P3
Vth[V]
Number of cells
(endurance
issue state)
E
VR1
P1
VR2
P2
VR3
P3
Vth[V]
Endurance
damage
issues

FIG. 4



# of Events
[a.u]
Limit #
Threshold #
Too short to defense with
general defense code
Case1
[abnormal]
Case2
[normal]
①
②
①
t1
t2
t3
Time[t]

FIG. 5

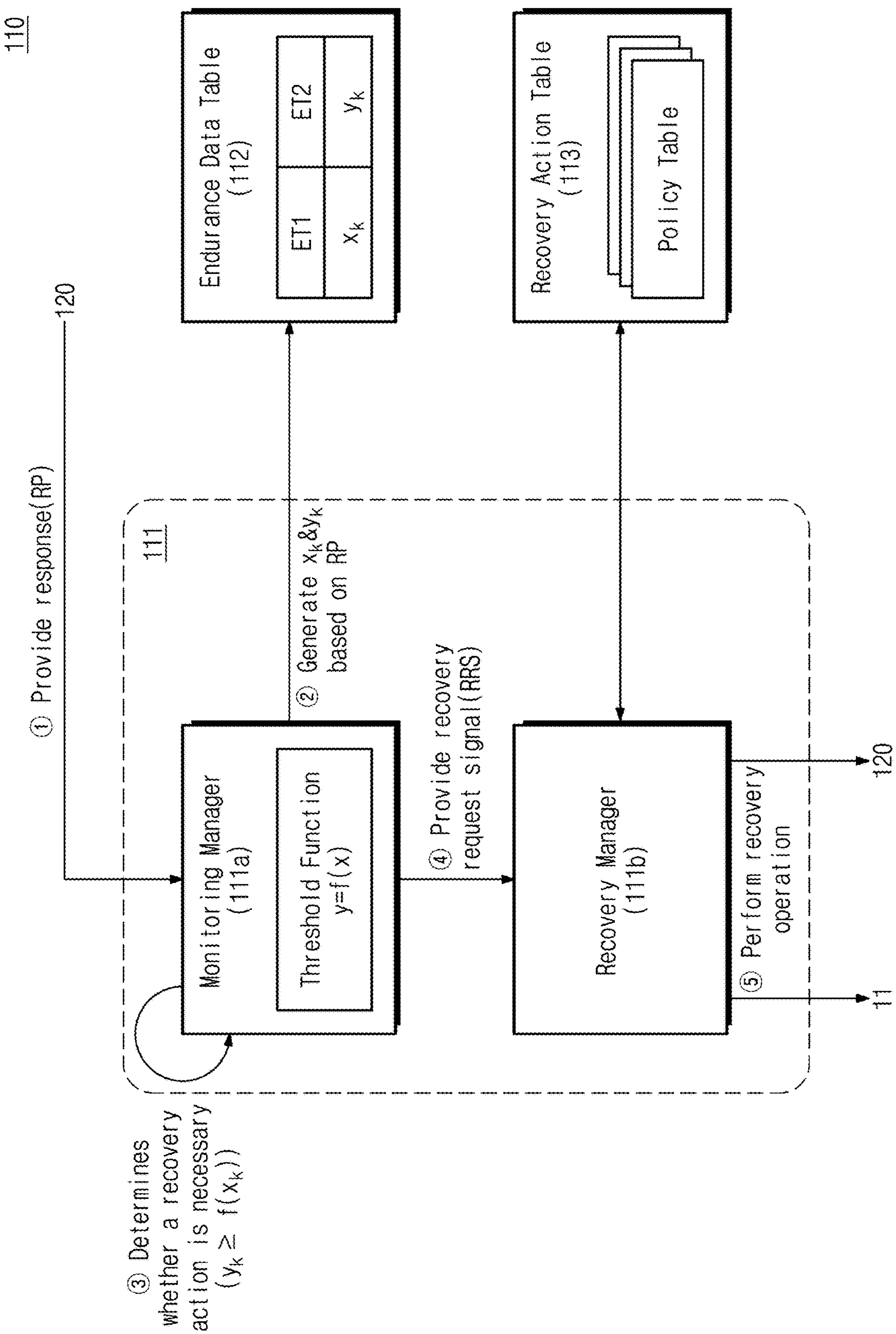

110
120
① Provide response(RP)
111
Monitoring Manager (111a)
Threshold Function y=f(x)
② Generate x_k&y_k based on RP
Endurance Data Table (112)
ET1
ET2
x_k
y_k
③ Determines whether a recovery action is necessary ($y_k \geq f(x_k)$)
④ Provide recovery request signal(RRS)
Recovery Manager (111b)
Recovery Action Table (113)
Policy Table
⑤ Perform recovery operation
120
11

FIG. 6

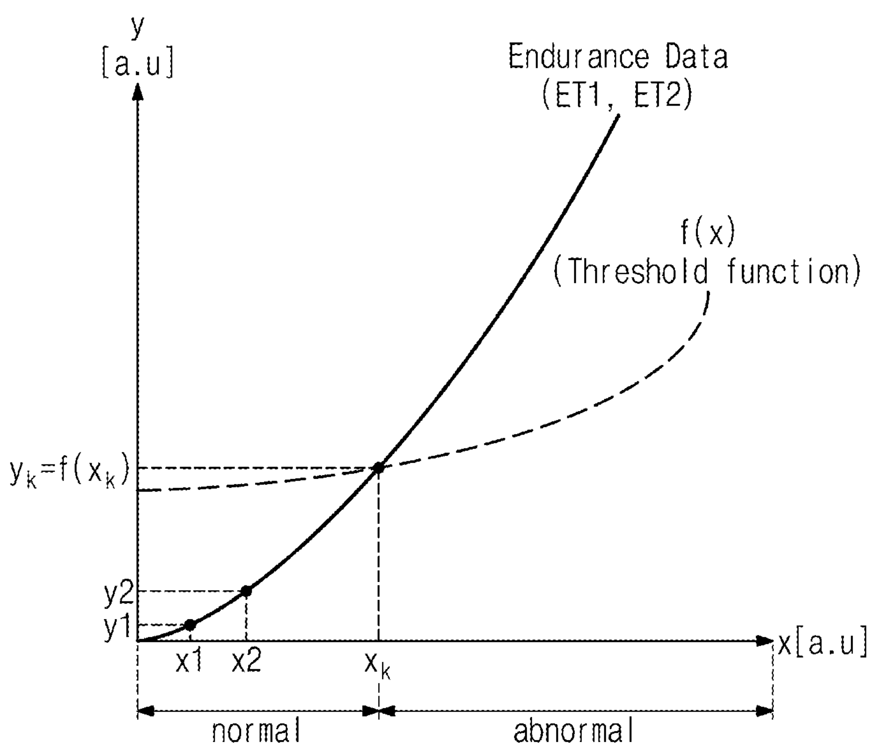

y
[a.u]
Endurance Data
(ET1, ET2)
f(x)
(Threshold function)
$y_k=f(x_k)$
y2
y1
x1
x2
x_k
x[a.u]
normal
abnormal

FIG. 7

Endurance Data		Threshold Function		Decision results
ET1	ET2	x	f(x)	
x1	y1	x1	f(x1)	Normal(y1 < f(x1))
x2	y2	x2	f(x2)	Normal(y2 < f(x2))
...	...	...	...	...
x_k	y_k	x_k	$f(x_k)$	Abnormal($y_k \geq f(x_k)$)

FIG. 8A

1st Policy Table

Endurance Data		Risk Level ($y_k - f(x_k)$)	Recovery Action
ET1	ET2		
time	EC	2 (> Th1)	Operation stop & Report
time	EC	1 (otherwise)	Restrict operation (Read Only)

FIG. 8B

2nd Policy Table

Endurance Data		Risk Level $(y_k - f(x_k))$	Recovery Action
ET1	ET2		
time	RC	2> (Th2)	Operation stop & Report
time	RC	1 (otherwise)	Reset read levels

FIG. 8C

3rd Policy Table

Endurance Data		Risk Level ($y_k - f(x_k)$)	Recovery Action
ET1	ET2		
EC or RC	Reclaim	2 (> Th3)	Operation stop & Report
EC or RC	Reclaim	1 (otherwise)	Restrict operation (Read Only)

FIG. 8D

4th Policy Table

Endurance Data		Risk Level ($y_k - f(x_k)$)	Recovery Action
ET1	ET2		
EC or RC	P/E Fail	2 (> Th4)	Operation stop & Report
EC or RC	P/E Fail	1 (otherwise)	Restrict operation (Read Only)

FIG. 8E

5th Policy Table

Endurance Data		Risk Level ($y_k - f(x_k)$)	Recovery Action
ET1	ET2		
EC or RC	Error Correct	2 (> Th5)	Operation stop & Report
EC or RC	Error Correct	1 (otherwise)	Restrict operation (Read Only)

FIG. 8F

6th Policy Table

Endurance Data		Risk Level ($y_k - f(x_k)$)	Recovery Action
ET1	ET2		
EC or RC	RTBB	2 (> Th6)	Operation stop & Report
EC or RC	RTBB	1 (otherwise)	Restrict operation (Read Only)

FIG. 9

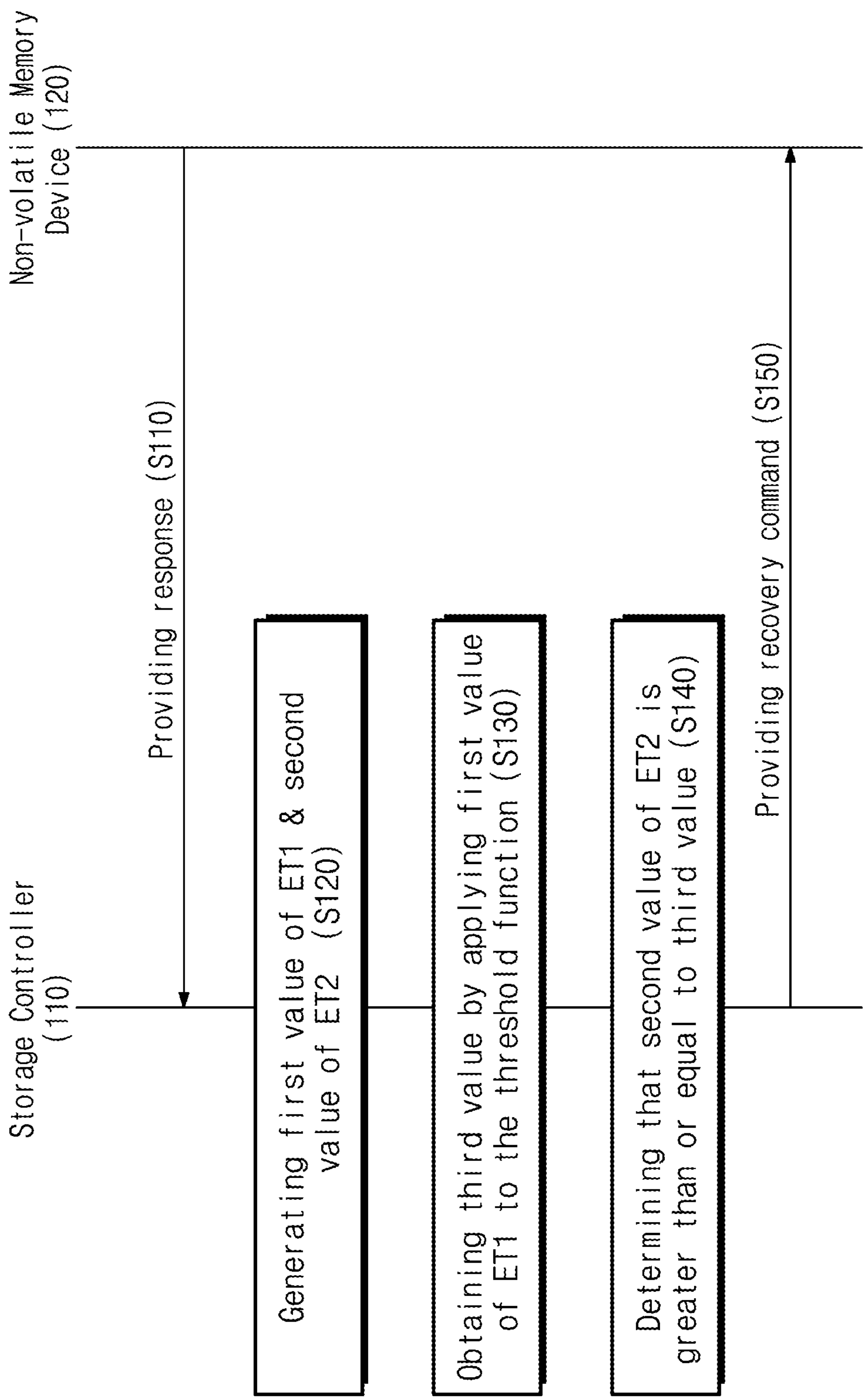

Storage Controller (110)
Non-volatile Memory Device (120)
Providing response (S110)
Generating first value of ET1 & second value of ET2 (S120)
Obtaining third value by applying first value of ET1 to the threshold function (S130)
Determining that second value of ET2 is greater than or equal to third value (S140)
Providing recovery command (S150)

FIG. 10

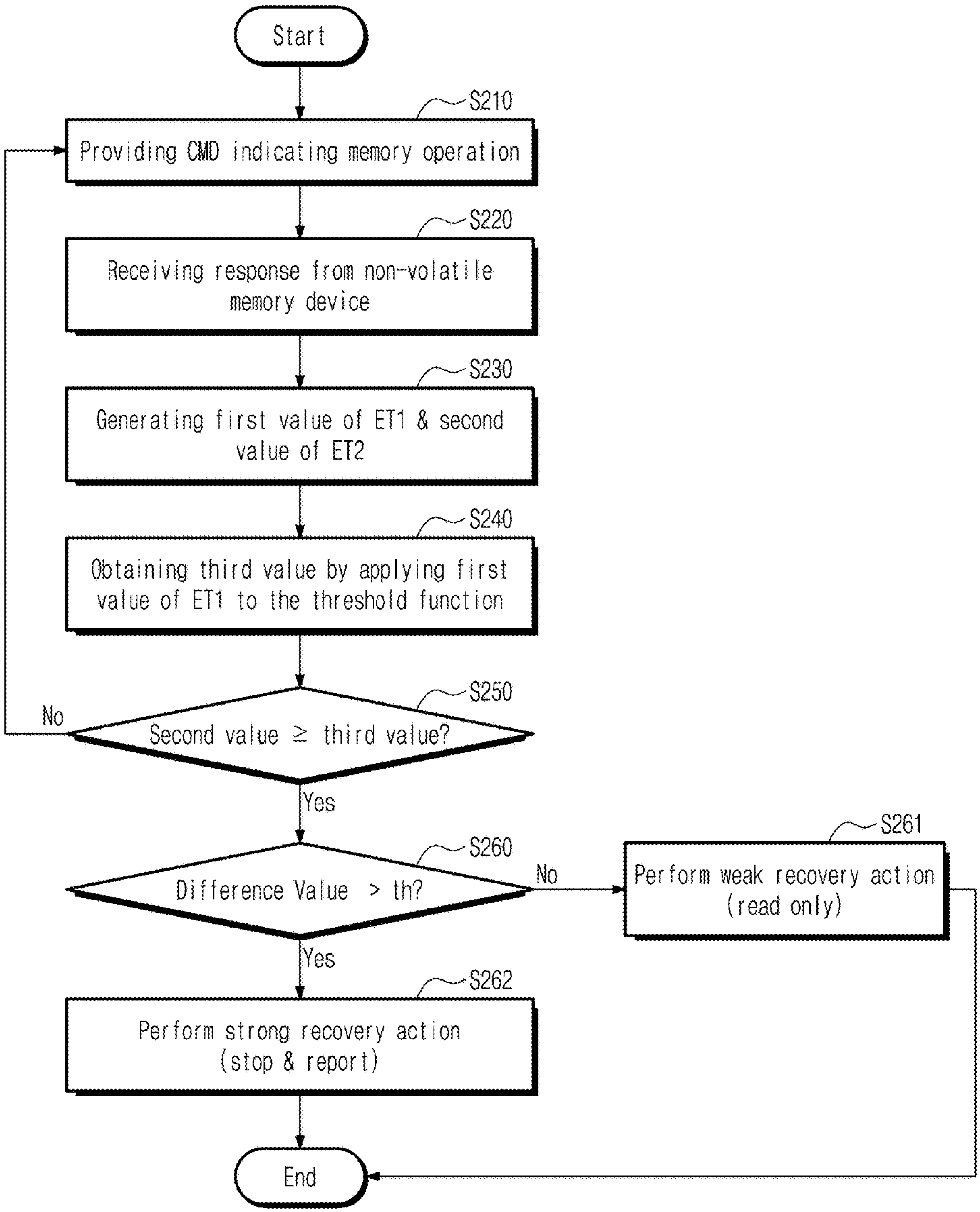
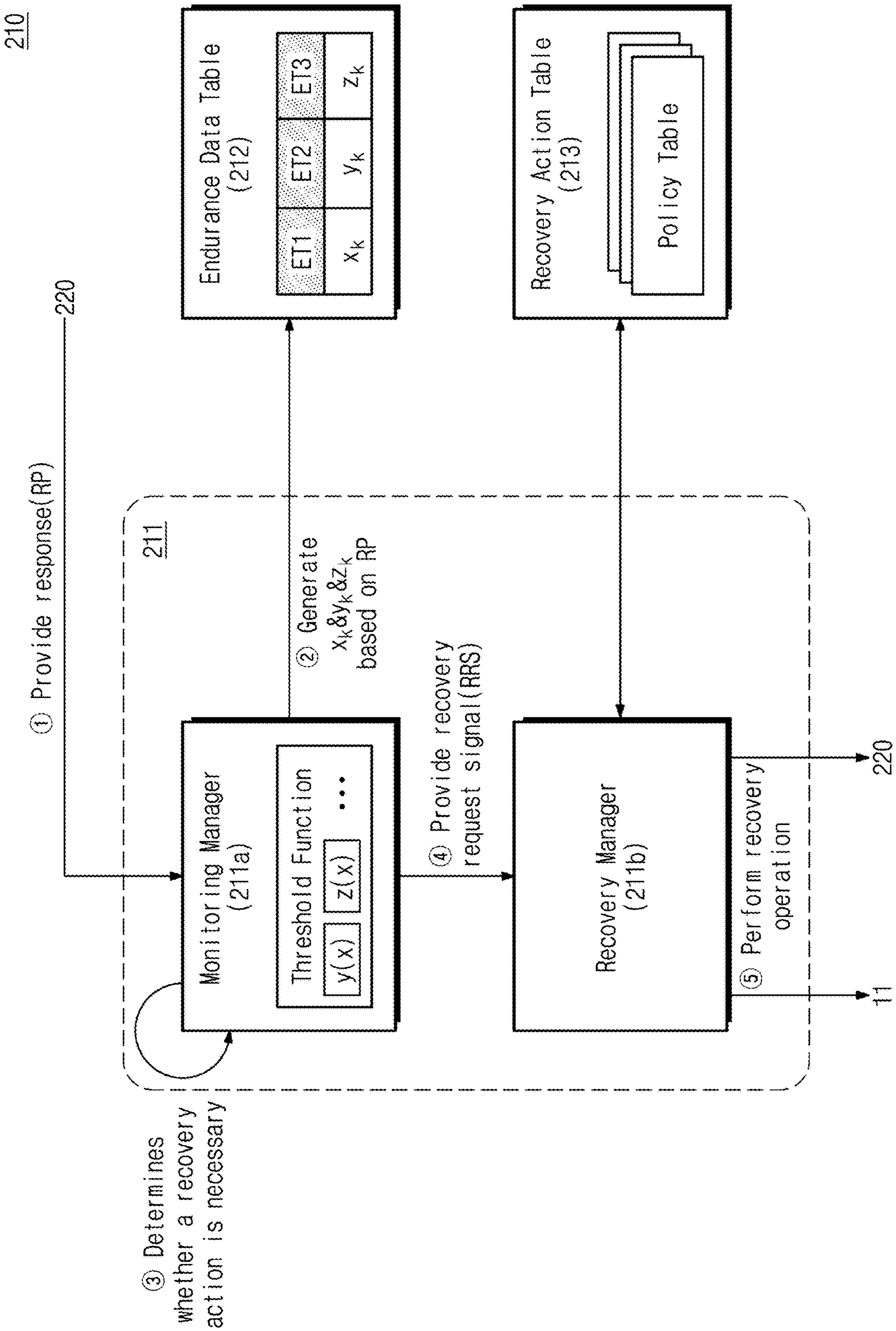

Start
S210
Providing CMD indicating memory operation
S220
Receiving response from non-volatile memory device
S230
Generating first value of ET1 & second value of ET2
S240
Obtaining third value by applying first value of ET1 to the threshold function
S250
Second value ≥ third value?
No
Yes
S260
Difference Value > th?
No
S261
Perform weak recovery action (read only)
Yes
S262
Perform strong recovery action (stop & report)
End

FIG. 11



210
220
① Provide response(RP)
211
Monitoring Manager (211a)
Threshold Function
y(x)
z(x)
...
② Generate x_k&y_k&z_k based on RP
③ Determines whether a recovery action is necessary
④ Provide recovery request signal(RRS)
Recovery Manager (211b)
⑤ Perform recovery operation
11
220
Endurance Data Table (212)
ET1
ET2
ET3
x_k
y_k
z_k
Recovery Action Table (213)
Policy Table

STORAGE DEVICE PERFORMING RECOVERY OPERATION BY USING ENDURANCE DATA, AND OPERATION METHOD THEREOF

CROSS-REFERENCE TO RELATED APPLICATION

[0001] This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2024-0025615 filed on Feb. 22, 2024, in the Korean Intellectual Property Office, the disclosure of which is herein incorporated by reference in its entirety.

BACKGROUND

[0002] Example embodiments of the present disclosure described herein relate to a storage device, and more particularly, relate to a storage device performing a recovery operation by using endurance data and an operation method thereof.

[0003] A memory device stores data in response to a write request and outputs data stored therein in response to a read request. For example, the memory device is classified as a volatile memory device, which loses data stored therein when a power is turned off, such as a dynamic random access memory (DRAM) device or a static RAM (SRAM) device, or a non-volatile memory device, which retains data stored therein even when a power is turned off, such as a flash memory device, a phase-change RAM (PRAM), a magnetic RAM (MRAM), or a resistive RAM (RRAM).

[0004] The endurance of the non-volatile memory device may be damaged due to various factors such as an increase in an erase count and a read count by a threshold voltage distribution change and an increase in a number of bad blocks. There is a need for a method of capable of preemptively preventing a damage to endurance of the non-volatile memory when the erase count, the read count, etc. increase during a short time.

SUMMARY

[0005] Example embodiments of the present disclosure provide a storage device performing a recovery operation by using endurance data and an operation method thereof.

[0006] According to an embodiment, an operating method of a storage controller configured to communicate with a non-volatile memory device includes receiving a memory response from a non-volatile memory device; generating a first value of a first endurance type and a second value of a second endurance type, based on the memory response, the first endurance type and the second endurance type being related to an endurance of the non-volatile memory device; obtaining a third value by applying the first value to a threshold function, the threshold function being defined based on a correlation between the first endurance type and the second endurance type; determining whether the second value is greater than or equal to the third value; and performing a recovery operation in response to determining that the second value is greater than or equal to the third value.

[0007] According to an embodiment, an operating method of a storage controller configured to communicate with a non-volatile memory device includes receiving a memory response from a non-volatile memory device; generating a first value of a first endurance type, a second value of a second endurance type, and a third value of a third endurance type, based on the memory response, the first endurance type, the second endurance type, and the third endurance type being related to an endurance of the non-volatile memory device; obtaining a fourth value by applying the first value to a first threshold function, wherein the first threshold function is defined based on a correlation between the first endurance type and the second endurance type;, obtaining a fifth value by applying the first value to a second threshold function, wherein the second threshold function is defined based on a correlation between the first endurance type and the third endurance type; and performing a recovery operation, based on at least one of determining that the second value is greater than or equal to the fourth value or determining that the third value is greater than or equal to the fifth value.

[0008] According to an embodiment, a storage device includes a non-volatile memory device, and a storage controller. The storage controller includes a monitoring manager configured to receive a memory response from a non-volatile memory device, generate a first value of a first endurance type and a second value of a second endurance type based on the memory response, obtain a third value by applying the first value to a threshold function, and generate a recovery request signal in response to determining that the second value is greater than or equal to the third value; and a recovery manager configured to perform a recovery operation in response to the recovery request signal. The first endurance type and the second endurance type may be related to an endurance of the non-volatile memory device, and the threshold function may be defined based on a correlation between the first endurance type and the second endurance type.

BRIEF DESCRIPTION OF DRAWINGS

[0009] The above and other objects and features of the present disclosure will become apparent by describing in detail certain example embodiments thereof with reference to the accompanying drawings.

[0010] FIG. **1** is a block diagram of a storage device according to one or more example embodiments of the present disclosure.

[0011] FIG. **2** is a block diagram illustrating a storage controller of FIG. **1** in detail.

[0012] FIG. **3** is a diagram illustrating threshold voltage distributions of related art multi-level cells.

[0013] FIG. **4** is a diagram describing an endurance defense code of a related art storage system.

[0014] FIG. **5** is a diagram describing an operating method of a storage controller according to one or more example embodiments of the present disclosure.

[0015] FIG. **6** is a diagram describing a threshold function and endurance data according to one or more example embodiments of the present disclosure.

[0016] FIG. **7** is a table describing a threshold function and endurance data according to one or more example embodiments of the present disclosure.

[0017] FIGS. **8**A to **8**F are tables illustrating a policy table of FIG. **5**.

[0018] FIG. **9** is a flowchart describing an operating method of a storage device according to an embodiment of the present disclosure.

[0019] FIG. **10** is a flowchart describing an operating method of a storage controller according to one or more example embodiments of present disclosure.

[0020] FIG. **11** is a diagram describing an operating method of a storage controller to monitor three or more endurance types in detail, according to one or more example embodiments of the present disclosure.

DETAILED DESCRIPTION

[0021] Below, example embodiments of the present disclosure will be described in detail and clearly to such an extent that one skilled in the art carries out example embodiments of the present disclosure easily.

[0022] FIG. **1** is a block diagram of a storage system according to an embodiment of the present disclosure. Referring to FIG. **1**, a storage system **10** may include a host **11** and a storage device **100**. In some embodiments, the storage system **10** may refer to a computing system, which is configured to process a variety of information, such as a personal computer (PC), a notebook, a laptop, a server, a workstation, a tablet PC, a smartphone, a digital camera, and a black box.

[0023] The host **11** may control all the operations of the storage system **10**. For example, the host **11** may store data in the storage device **100** or may read data stored in the storage device **100**.

[0024] The storage device **100** may include a storage controller **110** and a non-volatile memory device **120**. The non-volatile memory device **120** may store data. The storage controller **110** may store data in the non-volatile memory device **120** or may read data stored in the non-volatile memory device **120**. The non-volatile memory device **120** may operate under control of the storage controller **110**. For example, based on a command CMD indicating an operation and an address ADD indicating a location of data, the storage controller **110** may store the data in the non-volatile memory device **120** or may read the data stored in the non-volatile memory device **120**.

[0025] The non-volatile memory device **120** may include a plurality of memory blocks BLK. Each of the plurality of memory blocks BLK may include a plurality of memory cells. The plurality of memory cells may store data.

[0026] In some embodiments, the non-volatile memory device **120** may be a NAND flash memory device, but the present disclosure is not limited thereto. For example, the non-volatile memory device **120** may be one of various storage devices, which retain data stored therein even when a power is turned off, such as a phase-change random access memory (PRAM), a magnetic random access memory (MRAM), a resistive random access memory (RRAM), and a ferroelectric random access memory (FRAM).

[0027] The storage controller **110** may include an endurance manager **111**, an endurance data table **112**, and a recovery action table **113**.

[0028] The endurance manager **111** may generate endurance data by monitoring endurance of the non-volatile memory device **120** and may perform a recovery operation.

[0029] The endurance manager **111** may monitor the non-volatile memory device **120** to receive a memory response. The memory response may indicate a result of a memory operation performed by the non-volatile memory device **120**. For example, the memory operation may indicate at least one of a read operation of reading data stored in the non-volatile memory device **120**, an erase operation of deleting data stored in the non-volatile memory device **120**, a write operation of storing data in the non-volatile memory device **120**, etc. The memory response may include the read data corresponding to the read operation, information about whether a program corresponding to the write operation succeeds, information about whether an erase corresponding to the erase operation succeeds, etc.

[0030] The endurance manager **111** may generate the endurance data by using the memory response. The endurance data may include a plurality of endurance types and a value of each of the plurality of endurance types. Each of the plurality of endurance types may indicate a variable (or event) affecting the endurance or a variable affected by the endurance. The endurance may indicate, for example, how many program/erase cycles the non-volatile memory device **120** may endure. The endurance may be associated with a lifetime of the non-volatile memory device **120**.

[0031] In some embodiments, an endurance type may be one of the following for each memory block: an erase count, a read count, a program count, a reclaim count, a number of times of execution of an error correction code, an error correction success step, a number of bad blocks, a program fail count, an erase fail count, and a time. However, the present disclosure is not limited thereto. For example, kinds of the endurance type may increase or decrease depending on settings of the user or a design.

[0032] A reclaim may refer to an operation of again writing data stored in memory cells of the non-volatile memory device **120** in any other memory cells to guarantee reliability of data. For example, the storage controller **110** may perform a read operation on target data of a first memory block determined as requiring a read reclaim, may perform a write operation of writing the target data in a second memory block, and may perform an erase operation on the first memory block. In this case, the read operation may be based on an optimized read voltage level.

[0033] The error correction code (ECC) may refer to an algorithm which detects an error present in the read data read from the non-volatile memory device **120** and corrects the error. For example, an error may occur in the read data when threshold voltage distributions of the memory cells of the non-volatile memory device **120** change or when a leakage current occurs. The number of times of execution of the error correction code may indicate a number of times an error is determined to be present in the read data and the error is corrected.

[0034] The error correction success step may indicate a step, in which the error correction succeeds, from among a plurality of recovery steps included in a recovery algorithm. The recovery algorithm may indicate an algorithm defined in advance to be executed by the storage system **10** when an uncorrectable error occurs. The recovery algorithm may sequentially perform the plurality of recovery steps, and when the error correction succeeds, the recovery algorithm may not perform the following (or the remaining) recovery steps. The uncorrectable error may occur when threshold voltage distributions of memory cells change.

[0035] The bad block may be a run-time bad block. The bad block refers to a memory block which is determined as being impossible to perform the normal operation. The run-time bad block among bad blocks means a bad block which occurs while the non-volatile memory device **120** operates, not a bad block which occurs in a process of manufacturing the non-volatile memory device **120**. For example, when an error is incapable of being corrected even by an error correction algorithm or the recovery algorithm, the storage controller **110** may classify the corresponding memory block as a bad block and may not perform an operation on the corresponding memory block any more.

[0036] The program failure may indicate that the program operation performed by the non-volatile memory device **120** fails. In this case, the non-volatile memory device **120** may provide the storage controller **110** with the memory response indicating the failure of the program operation.

[0037] The erase failure may indicate that the erase operation performed by the non-volatile memory device **120** fails. In this case, the non-volatile memory device **120** may provide the storage controller **110** with the memory response indicating the failure of the erase operation.

[0038] The time may indicate a system time (or a system clock) which is used in the storage system **10**.

[0039] The recovery operation may indicate a preventive operation or a recovery operation for suppressing the reduction of endurance of the non-volatile memory device **120**. The reduction of endurance of the non-volatile memory device **120** may be caused by an abnormal increase in the erase count or the read count. A phenomenon such as the abnormal increase in the erase count or the read count may be caused by a setting error of a storage device, a change in threshold voltage distributions of memory cells, etc.

[0040] In some embodiments, the recovery operation may be an operation of limiting the non-volatile memory device **120** such that only a restricted operation (e.g., a read operation) is performed or an operation of reporting a situation to the host **11** in a state where the operation of a storage controller stops.

[0041] The endurance manager **111** may determine the recovery operation in consideration of the endurance data and at least one threshold function. The threshold function may be a function which is defined based on a correlation between two endurance types among the plurality of endurance types. The threshold function may be defined in advance.

[0042] In some embodiments, the threshold function may indicate an expected correlation between two different endurance types among the plurality of endurance types of the storage system **10**. The expected correlation may indicate a correlation between two different endurance types which a normal storage system **10** (or the storage system **10** whose endurance is not damaged) is expected to have.

[0043] In other words, the threshold function may indicate a tendency associated with the endurance types of the storage system **10**. For example, the threshold function may indicate the tendency of the increase in the reclaim count according to the erase count, or the number of times of execution of an error correction code according to the erase count.

[0044] The threshold function may be a function determined in advance. For example, the threshold function may be determined based on a machine learning model trained by using a large amount of data sets associated with the plurality of endurance types. The endurance manager **111** may store at least one threshold function.

[0045] The endurance manager **111** may determine the recovery operation in consideration of the endurance data and at least one threshold function corresponding to the endurance data.

[0046] First, the endurance manager **111** may determine whether the recovery operation is required, based on a first comparison operation of the endurance data and at least one threshold function corresponding to the endurance data.

[0047] For example, when the endurance data include a first value of a first endurance type and a second value of a second endurance type, a first threshold function corresponding to the endurance data is defined based on a correlation between the first endurance type and the second endurance type. The endurance manager **111** may obtain a third value by applying the first value of the first threshold function. The endurance manager **111** may compare the second value of the second endurance type and the third value, and when the second value is greater than or equal to the third value, the endurance manager **111** may determine that the recovery operation is required.

[0048] The endurance manager **111** may determine the recovery operation based on a first comparison result of the first comparison operation. For example, the first comparison result may include a difference value between the second value of the second endurance type and the third value obtained from the first threshold function. As the difference value becomes greater, the risk that the endurance of the non-volatile memory device **120** is damaged may become higher.

[0049] The endurance manager **111** may determine the recovery operation based on a second comparison operation of the difference value of the first comparison operation and a threshold value. The threshold value which is a value determined in advance may be determined depending on the first and second endurance types.

[0050] For example, when the difference value of the first comparison result is greater than the threshold value, the endurance manager **111** may determine to perform a strong recovery action as the recovery operation. The endurance manager **111** may stop all operations of the non-volatile memory device **120** and may provide a report to the host **11**. In contrast, when the difference value of the first comparison result is smaller than or equal to the threshold value, the endurance manager **111** may determine to perform a weak recovery action as the recovery operation. The endurance manager **111** may manage the non-volatile memory device **120** such that only a restricted operation (e.g., a read operation) is performed.

[0051] The endurance data table **112** may include the plurality of endurance types and the values of the plurality of endurance types. The endurance data table **112** may be updated by the endurance manager **111**.

[0052] In some embodiments, when a response is received from the non-volatile memory device **120**, the endurance manager **111** may generate the endurance data based on the received response by referring to the endurance data table **112**.

[0053] For example, the endurance manager **111** may obtain existing endurance data by referring to the endurance data table **112**. The existing endurance data may indicate the endurance data of the non-volatile memory device **120** before at least one operation corresponding to the response is performed. The endurance manager **111** may update the existing endurance data, based on the response received from the non-volatile memory device **120**. The updated endurance data may correspond to the above endurance data, that is, the endurance data that is used to determine whether the recovery operation is required.

[0054] In detail, when the response received from the non-volatile memory device **120** corresponds to one or more endurance types, the endurance manager **111** may update a value of each of the one or more endurance types corresponding to the response. For example, when the response corresponds to a first endurance type having a value of "n" and corresponds to a second endurance type having a value of "m", the endurance manager **111** may generate a value obtained by adding "n" to a first existing value of the first endurance type included in the existing endurance data as a first value of the first endurance type of the updated endurance data. The endurance manager **111** may generate a value obtained by adding "m" to a second existing value of the second endurance type included in the existing endurance data as a second value of the second endurance type of the updated endurance data.

[0055] In this case, the endurance manager **111** may generate a value of each of endurance types, whose values are not changed in the existing endurance data, as a value of the endurance data without modification.

[0056] The endurance manager **111** may update the endurance data table **112**, based on the updated endurance data.

[0057] The recovery action table **113** may include a plurality of policy tables. Each of the plurality of policy tables may correspond to a combination of two different endurance types. For example, a first policy table may correspond to a combination of a first endurance type and a second endurance type, and a second policy table may correspond to a combination of the second endurance type and a third endurance type.

[0058] Each of the plurality of policy tables may include a combination of corresponding endurance types, a plurality of risk levels, and a plurality of recovery actions respectively corresponding to the plurality of risk levels. Each of the plurality of risk levels may have different risk level values depending on a degree to which the second value of the second endurance type deviates from a value obtained according to the threshold function. For example, each of the plurality of risk levels may indicate one of a first risk level and a second risk level.

[0059] The endurance manager **111** may compare the difference value and the threshold value (e.g., a target threshold value) and may determine a target risk level among the plurality of risk levels. The difference value may indicate a difference of the second value of the second endurance type and the third value obtained by applying the first value to the threshold function. The threshold value may be determined based on a combination of the first endurance type and the second endurance type.

[0060] In some embodiments, the endurance manager **111** may store a threshold value table. The threshold value table may include at least one combination of two different endurance types belonging to each endurance type group and a plurality of threshold values each corresponding to the at least one combination. The endurance manager **111** may obtain the target threshold value by referring to the threshold value table based on the combination of the first endurance type and the second endurance type.

[0061] The plurality of recovery actions may include the strong recovery action and the weak recovery action.

[0062] In some embodiments, the recovery action table **113** may include the plurality of policy tables. Each of the plurality of policy tables may include a plurality of risk levels and a plurality of recovery actions, which are associated with two endurance types among the plurality of endurance types. For example, a first policy table among the plurality of policy tables may include a plurality of risk levels and a plurality of recovery actions, which are associated with the first endurance type (e.g., a time) and the second endurance type (e.g., the erase count), respectively. A second policy table among the plurality of policy tables may include a plurality of risk levels and a plurality of recovery actions, which are associated with the second endurance type (e.g., the erase count) and the third endurance type (e.g., the reclaim count), respectively. This will be described in detail with reference to FIGS. **8**A to **8**F.

[0063] For convenience of description, the description is given using an example in which each of the plurality of recovery actions is one of the strong recovery action and the weak recovery action, but the present disclosure is not limited thereto. For example, it may be possible to classify each of the plurality of recovery actions as one of three or more recovery action kinds by comparing the above difference value with a plurality of threshold values.

[0064] The endurance manager **111** may determine a target recovery action among the plurality of recovery actions by referring to the recovery action table **113** based on the target risk level.

[0065] The endurance manager **111** may perform the determined target recovery action as the recovery operation. For example, the endurance manager **111** may control the non-volatile memory device **120** such that the restricted operation is performed; alternatively, the endurance manager **111** may stop all operations of the non-volatile memory device **120** and may provide a report to the host **11**. Accordingly, the host (or user) may preserve (e.g., back up) the data stored in the non-volatile memory device **120** before the non-volatile memory device **120** is broken down.

[0066] The storage device **100** according to an embodiment of the present disclosure may perform the recovery operation whenever an endurance-related event occurs in the non-volatile memory device **120** or the tendency of occurrence of the endurance-related event is out of a normal tendency, that is, when a probability that the endurance is to be damaged is high. Accordingly, the endurance damage may be preemptively prevented without a separate request from the host.

[0067] Also, because the recovery operation is differently performed depending on the degree to which the tendency of occurrence of the endurance-related event is out of the normal tendency, the storage device **100** may have improved endurance coinciding with demands of various storage devices.

[0068] FIG. **2** is a block diagram illustrating a storage controller of FIG. **1** in detail. Referring to FIGS. **1** and **2**, the storage controller **110** may communicate with the host **11** and the non-volatile memory device **120**. The storage controller **110** may include the endurance manager **111**, the endurance data table **112**, the recovery action table **113**, a volatile memory device **114**, a processor **115**, a read only memory (ROM) **116**, an error correcting code (ECC) engine **117**, a host interface circuit **118**, and a non-volatile memory interface circuit **119**.

[0069] The endurance manager **111** may generate the endurance data based on a response received from the non-volatile memory device **120** and may perform the recovery operation based on the endurance data. The endurance data table **112** may include the plurality of endurance types and a plurality of values respectively corresponding to the plurality of endurance types. The recovery action table **113** may include the plurality of endurance types, the plurality of risk levels, and the plurality of recovery actions in association with one another.

[0070] In some embodiments, the endurance manager **111**, the endurance data table **112**, and the recovery action table **113** may be implemented with a firmware module. For example, the processor **115** may implement the endurance manager **111**, the endurance data table **112**, and the recovery action table **113** by loading instructions stored in the non-volatile memory device **120** to the volatile memory device **114** and executing the loaded instructions. However, the present disclosure is not limited thereto. For example, the endurance manager **111**, the endurance data table **112**, and the recovery action table **113** may be implemented with separate hardware or may be implemented with a combination of hardware and software.

[0071] The volatile memory device **114** may be used as a main memory, a buffer memory, or a cache memory of the storage controller **110**. The processor **115** may control an overall operation of the storage controller **110**. The ROM **116** may be used as a read only memory which stores information necessary for the operation of the storage controller **110**.

[0072] The ECC engine **117** may detect and correct an error of data obtained from the non-volatile memory device **120**. For example, the ECC engine **117** may have an error correction capability of a given level. The ECC engine **117** may manage data having an error level (e.g., a number of flipped bits) exceeding the error correction capability as uncorrectable error data.

[0073] The storage controller **110** may communicate with the host **11** through the host interface circuit **118**. In some embodiments, the host interface circuit **118** may be implemented based on at least one of various interfaces such as a serial ATA (SATA) interface, a peripheral component interconnect express (PCIe) interface, a serial attached SCSI (SAS), a non-volatile memory express (NVMe) interface, and a universal flash storage (UFS) interface.

[0074] The storage controller **110** may communicate with the non-volatile memory device **120** through the non-volatile memory interface circuit **119**. In some embodiments, the non-volatile memory interface circuit **119** may be implemented based on a NAND interface.

[0075] FIG. **3** is a diagram illustrating threshold voltage distributions of related art multi-level cells. A graph showing threshold voltage distributions of multi-level cells MLC each storing two bits is illustrated in FIG. **3**. Below, for convenience of description, the multi-level cell MLC is intended to refer to a memory cell storing two bits, a memory cell storing three bits is referred to as a "triple level cell TLC", and a memory cell storing four bits is referred to as a "quadruple level cell QLC".

[0076] Referring to the graph of the multi-level cells MLC, the horizontal axis represents a threshold voltage (e.g., a level of a threshold voltage), and the vertical axis represents a number of memory cells. The multi-level cell MLC may have one of an erase state "E" and first to third programming states P**1**, P**2**, and P**3** whose threshold voltage distributions sequentially increase.

[0077] In the multi-level cell MLC, a first read voltage level VR**1***i* may refer to a voltage for distinguishing the erase state "E" from the first programming state P**1**. A second read voltage level VR**2***i* may refer to a voltage for distinguishing the first programming state P**1** from the second programming state P**2**. A third read voltage level VR**3***i* may refer to a voltage for distinguishing the second programming state P**2** from the third programming state P**3**. Like the graph of FIG. **3**, in the multi-level cells MLC, it would be understood that the erase state "E" and the first to third programming states P**1**, P**2**, and P**3** are distinguished by the first to third read voltage levels VR**1***i*, VR**2***i*, and VR**3***i* without overlapping each other.

[0078] However, the threshold voltage distributions of the multi-level cells MLC may change due to various causes; in this case, the first to third programming states P**1**, P**2**, and P**3** may partially overlap each other. For example, as the write operation and the erase operation on the multi-level cells MLC are repeated over time, the threshold voltage distributions may change. When the threshold voltage distributions change, in some cases, there may be a need to determine new first to third read voltage levels VR**1**, VR**2**, and VR**3** capable of distinguishing the erase state "E" and the first to third programming states P**1**, P**2**, and P**3**.

[0079] To solve the issue that an error continuously occurs as the non-volatile memory device **120** performs the memory operation in a state where the threshold voltage distributions of the multi-level cells MLC changes, other memory operations (e.g., a reclaim operation) may be performed, causing the damage of endurance of the non-volatile memory device **120**. A related art storage system may execute an endurance defense code to prevent the damage of endurance.

[0080] FIG. **4** is a diagram describing an endurance defense code of a related art storage system. Referring to FIG. **4**, points in time when a related art storage system executes the endurance defense code in a normal state and an abnormal state are illustrated. In FIG. **4**, the horizontal axis represents a time, and the vertical axis represents the number of times of occurrence of an endurance-related event. The endurance-related event which is the endurance type described with reference to FIG. **1** may indicate one of the erase count, the read count, the reclaim count, the number of bad blocks, and the number of times of error correction. The number of endurance-related events may increase as a time passes.

[0081] A limit value of the vertical axis indicates a number of times of occurrence of an endurance-related event, which the storage system is capable of enduring. In other words, the number of times of occurrence of an event exceeds the limit value, the storage system may not operate any more.

[0082] A threshold value of the vertical axis indicates the number of times of occurrence of an event, at which the endurance defense code of the related art storage system is executed. In other words, when the number of times of occurrence of an endurance-related event reaches a threshold value, the related art storage system executes the endurance defense code. The endurance defense code may indicate an algorithm which a storage controller executes to prevent the damage of endurance of a non-volatile memory device.

[0083] For example, when the number of times of occurrence of an endurance-related event increases and then reaches the threshold value at a third point in time t**3**, the related art storage system may execute the endurance defense code at the third point in time t**3** such that the number of times of occurrence of an endurance-related event does not reach the limit value.

[0084] However, it is difficult for the related art storage system to defend against the damage of endurance in an abnormal situation where the number of times of occurrence of an endurance-related event sharply increases over time. For example, when the number of times of occurrence of an endurance-related event sharply increases over time and then reaches the threshold value at the first point in time t**1**, the related art storage system may execute the endurance defense code at the first point in time t**1**. However, when the number of times of occurrence of an endurance-related event reaches the limit value at a second point in time t**2** due to the sharp increase before the effect of the endurance defense code appears, the storage system may be broken out.

[0085] FIG. **5** is a diagram describing an operating method of a storage controller according to one or more example embodiments of the present disclosure. Referring to FIGS. **1** and **5**, the storage controller **110** may communicate with the host **11** and the non-volatile memory device **120**. The storage controller **110** may include the endurance manager **111**, the endurance data table **112**, and the recovery action table **113**.

[0086] The endurance manager **111** may include a monitoring manager **111***a* and a recovery manager **111***b*.

[0087] The monitoring manager **111***a* may monitor the non-volatile memory device **120** to generate the endurance data. The endurance data may include a first endurance type ET**1**, a first value x_k of the first endurance type ET**1**, a second endurance type ET**2**, and a second value y_k of the second endurance type ET**2**.

[0088] The monitoring manager **111***a* may receive a response RP from the non-volatile memory device **120**. The monitoring manager **111***a* may obtain existing endurance data by referring to the endurance data table **112**, based on receiving the response RP. The monitoring manager **111***a* may generate the endurance data by updating the existing endurance data based on the response RP. The monitoring manager **111***a* may update the endurance data table **112** based on the updated endurance data.

[0089] The monitoring manager **111***a* may store a threshold function "y=f(x)". The threshold function fx may be defined based on a correlation between the first endurance type ET**1** and the second endurance type ET**2**. For example, the threshold function fx may define a value of the second endurance type ET**2**, which is expected depending on a value of the first endurance type ET**1**. The monitoring manager **111***a* may obtain a first expected value $f(x_k)$ by applying the first value x_k of the first endurance type ET**1** to the threshold function fx. The monitoring manager **111***a* may perform the first comparison operation of comparing the first expected value $f(x_k)$ and the second value y_k of the second endurance type ET**2**.

[0090] The monitoring manager **111***a* may generate a recovery request signal RRS, based on determining that the second value y_k is greater than or equal to the first expected value $f(x_k)$. The recovery request signal RRS may be a signal indicating an initiation of the recovery operation of the recovery manager **111***b*. The recovery request signal RRS may include the first endurance type ET**1**, the second endurance type ET**2**, and a difference value of the second value y_k and the first expected value $f(x_k)$.

[0091] The monitoring manager **111***a* may provide the recovery request signal RRS to the recovery manager **111***b*.

[0092] The recovery manager **111***b* may receive the recovery request signal RRS from the monitoring manager **111***a*. The recovery manager **111***b* may determine a recovery threshold value, based on the first endurance type ET**1** and the second endurance type ET**2** of the recovery request signal RRS. The recovery threshold value may be defined based on the first endurance type ET**1** and the second endurance type ET**2** and may be determined in advance.

[0093] In some embodiments, the recovery manager **111***b* may include a recovery threshold value table. The recovery threshold value table may include at least one combination of two different endurance types belonging to each endurance type group and a plurality of recovery threshold values each corresponding to the combination. The recovery manager **111***b* may determine a recovery threshold value corresponding to the first endurance type ET**1** and the second endurance type ET**2** by referring to the recovery threshold value table.

[0094] The recovery manager **111***b* may determine a risk level based on the second comparison operation of comparing the difference value and the determined recovery threshold value.

[0095] In some embodiments, the risk level may be one of a first level and a second level. For example, the recovery manager **111***b* may generate the risk level as the first level, based on determining that the difference value is smaller than or equal to the recovery threshold value. In contrast, the recovery manager **111***b* may generate the risk level as the second level, based on determining that the difference value is greater than the recovery threshold value.

[0096] The recovery manager **111***b* may determine a recovery action by referring to the recovery action table **113** based on the first endurance type ET**1**, the second endurance type ET**2**, and the risk level.

[0097] The recovery action table **113** may include at least one combination of two different endurance types belonging to each endurance type group, a plurality of risk levels, and a plurality of recovery actions corresponding to one of the combinations of the endurance type groups and one of the risk levels.

[0098] In some embodiments, the recovery action table **113** may include the plurality of policy tables. Each of the plurality of policy tables may correspond to one of the combinations of the endurance type groups each including two different endurance types. This will be described in detail with reference to FIGS. **8**A to **8**F.

[0099] The recovery manager **111***b* may perform the determined recovery action. For example, the recovery manager **111***b* may control the non-volatile memory device **120** such that only a restricted operation is performed; alternatively, the recovery manager **111***b* may stop an operation of the non-volatile memory device **120** and may provide a report to the host **11**.

[0100] Below, an operating method of the storage controller **110** according to one or more example embodiments of the present disclosure will be described in detail.

[0101] In a first operation ①, the monitoring manager **111***a* may receive the response RP from the non-volatile memory device **120**. The response RP may correspond to the memory operation of the non-volatile memory device **120**.

[0102] In a second operation ②, the monitoring manager **111***a* may generate the endurance data based on the response RP. The monitoring manager **111***a* may obtain existing endurance data by referring to the endurance data table **112**.

The monitoring manager **111***a* may generate the first value x_k of the first endurance type ET**1** and the second value y_k of the second endurance type ET**2** by updating the existing endurance data based on the response RP. The monitoring manager **111***a* may update the endurance data table **112** based on the first value x_k of the first endurance type ET**1** and the second value y_k of the second endurance type ET**2**.

[0103] In a third operation 3, the monitoring manager **111***a* may determine whether the recovery action is necessary. The monitoring manager **111***a* may determine that the recovery action is necessary, based on determining that the second value y_k is greater than or equal to the first expected value "$y=f(x_k)$" obtained by applying the first value x_k to the threshold function fx.

[0104] In a fourth operation 4, the monitoring manager **111***a* may provide the recovery request signal RRS to the recovery manager **111***b*.

[0105] In a fifth operation 5, the recovery manager **111***b* may perform the recovery operation. In detail, the recovery manager **111***b* may determine the recovery action by referring to the recovery action table **113** based on the recovery request signal RRS. For example, the recovery manager **111***b* may determine the recovery action based on the recovery threshold value, which is determined according to the recovery request signal RRS.

[0106] In some embodiments, the recovery manager **111***b* may allow the non-volatile memory device **120** to perform only the read operation, based on determining the weak recovery action as the recovery operation. In some embodiments, based on determining the strong recovery action as the recovery operation, the recovery manager **111***b* may stop an operation of the non-volatile memory device **120** and may notify the host **11** that the operation of the non-volatile memory device **120** is stopped.

[0107] FIG. **6** is a diagram describing a threshold function and endurance data according to one or more example embodiments of the present disclosure. The threshold function and the first comparison operation using the threshold function and the endurance data will be described with reference to FIGS. **5** and **6**. In FIG. **6**, the horizontal axis represents the first value "x" of the first endurance type ET**1**, and the vertical axis represents the second value "y" of the second endurance type ET**2**.

[0108] The threshold function fx may be defined in advance based on a correlation between the first endurance type ET**1** and the second endurance type ET**2**. The threshold function fx may be referred to as a "tendency of the second endurance type ET**2** to the first endurance type ET**1**".

[0109] An endurance manager may compare the second value "y" and a value obtained by applying the first value "x" to the threshold function fx and may determine whether the recovery operation is required.

[0110] The first value "x" and the second value "y" of the endurance data increases whenever a non-volatile memory device performs an operation corresponding to the endurance type. For example, when the first value "x" is a first X-axis value **x1**, the second value "y" may be a first Y-axis value **y1**; when the first value "x" is a second X-axis value **x2**, the second value "y" may be a second Y-axis value **y2**; when the first value "x" is a k-th X-axis value x_k, the second value "y" may be a k-th Y-axis value y_k.

[0111] When the first value "x" is the first X-axis value **x1** or the second X-axis value **x2**, the first Y-axis value **y1** or the second Y-axis value **y2** is smaller than a value obtained by applying the first value "x" to the threshold function fx. The endurance manager may determine the state of the non-volatile memory device as a normal state where a probability that the endurance of the non-volatile memory device is damaged is low and may determine that the recovery operation is not required.

[0112] When the first value "x" is the k-th X-axis value x_k, the k-th Y-axis value yk is equal to a value $f(x_k)$ obtained by applying the k-th X-axis value xx to the threshold function fx. In this case, the endurance manager may determine the state of the non-volatile memory device as an abnormal state where a probability that the endurance of the non-volatile memory device is damaged is high and may determine that the recovery operation is required.

[0113] In some embodiments, the first endurance type ET**1** may be a time, and the second endurance type ET**2** may indicate the erase count. In this case, the erase count may indicate the erase count before a first refresh operation is performed or may indicate the erase count before a next refresh operation is performed after the first refresh operation is performed. The endurance manager may use the threshold function fx indicating a tendency of the erase count over time.

[0114] In some embodiments, the first endurance type ET**1** may be a time, and the second endurance type ET**2** may indicate the read count. In this case, the read count may indicate the read count before the first refresh operation is performed or may indicate the read count before a next refresh operation is performed after the first refresh operation is performed. The endurance manager may use the threshold function fx indicating a tendency of the read count over time.

[0115] In some embodiments, the first endurance type ET**1** may indicate one of the erase count and the read count. The second endurance type ET**2** may be the reclaim count. The endurance manager may use the threshold function fx indicating a tendency of the reclaim count according to the erase count or the threshold function fx indicating a tendency of the reclaim count according to the read count.

[0116] In some embodiments, the first endurance type ET**1** may indicate one of the erase count and the read count. The second endurance type ET**2** may be the number of bad blocks. The endurance manager may use the threshold function fx indicating a tendency of the number of bad blocks according to the erase count or the threshold function fx indicating a tendency of the number of bad blocks according to the read count.

[0117] In some embodiments, the first endurance type ET**1** may indicate one of the erase count and the read count. The second endurance type ET**2** may indicate the number of times of execution of an error correction code. The endurance manager may use the threshold function fx indicating a tendency of the number of times of execution of an error correction code according to the erase count or the threshold function fx indicating a tendency of the number of times of execution of an error correction code according to the read count.

[0118] In some embodiments, the first endurance type ET**1** may indicate one of the erase count and the read count. The second endurance type ET**2** may indicate an error correction step. The endurance manager may use the threshold function fx indicating a tendency of an error correction step according to the erase count or the threshold function fx indicating a tendency of an error correction step according to the read count.

[0119] The storage device according to the present disclosure may prevent the damage of endurance by performing the recovery operation when the endurance data are out of the tendency expected for a normal storage device.

[0120] For convenience of description, the threshold function fx and the endurance data are illustrated as the second value "y" of the second endurance type ET**2** increases as the first value "x" of the first endurance type ET**1** increases, but the present disclosure is not limited thereto. Even if the first value "x" increases, the second value "y" may be maintained without increasing.

[0121] FIG. **7** is a table describing a threshold function and endurance data according to one or more example embodiments of the present disclosure. The first comparison operation of ocomparing the endurance data and the threshold function fx, performed the endurance manager **111**, will be described with reference to FIGS. **5** and **7**.

[0122] The endurance data may include a first value of the first endurance type ET**1** and a second value of the second endurance type ET**2**. The threshold function fx may be defined based on a correlation between the first endurance type ET**1** and the second endurance type ET**2**.

[0123] In some embodiments, the endurance manager may generate a determination result based on a comparison result of the first comparison operation. The determination result may indicate one of a normal state or an abnormal state. The endurance manager may perform the recovery operation based on determining that the determination result indicates the abnormal state and may not perform the recovery operation based on determining that the determination result indicates the normal state.

[0124] The first value and the second value may be a first X-axis value **x1** and a first Y-axis value **y1**, respectively. The endurance manager may compare the first Y-axis value **y1** with a value f(**x1**) obtained by applying the first X-axis value **x1** to the threshold function fx. When the first Y-axis value **y1** is smaller than the obtained value f(**x1**), the endurance manager may determine that the non-volatile memory device is in the normal state.

[0125] The first value and the second value may be a second X-axis value **x2** and a second Y-axis value **y2**, respectively. The endurance manager may compare the second Y-axis value **y2** with a value f(**x2**) obtained by applying the second X-axis value **x2** to the threshold function fx. When the second Y-axis value **y2** is smaller than the obtained value f(**x2**), the endurance manager may determine that the non-volatile memory device is in the normal state.

[0126] The first value and k value may be a k-th X-axis value x_k and a k-th Y-axis value y_k, respectively. The endurance manager may compare the k-th Y-axis value y_x with a value $f(x_k)$ obtained by applying the k-th X-axis value x_k to the threshold function fx. When the k-th Y-axis value y_k is greater than or equal than the obtained value $f(x_k)$, the endurance manager may determine that the non-volatile memory device is in the abnormal state. The endurance manager may perform the recovery operation.

[0127] FIGS. **8**A to **8**F are tables illustrating a policy table of FIG. **5**. For convenience of description, FIGS. **8**A to **8**F may be referred to as "FIG. **8**". Referring to FIGS. **5** and **8**, each of the plurality of policy tables of FIG. **5** is illustrated as an example. The policy table may include the first endurance type ET**1**, the second endurance type ET**2**, the plurality of risk levels, and the plurality of recovery actions respectively corresponding to the plurality of risk levels.

[0128] FIG. **8**A shows the policy table corresponding to a case where the first endurance type ET**1** is a time and the second endurance type ET**2** is an erase count EC.

[0129] When the risk level is the first level, that is, when the endurance manager determines that the difference value does not exceed a first threshold value Th**1**, the endurance manager may control the non-volatile memory device such that only a restricted operation is performed. For example, the endurance manager may allow the non-volatile memory device to perform only the read operation.

[0130] When the risk level is the second level, that is, when the endurance manager determines that the difference value exceeds the first threshold value Th**1**, the endurance manager may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0131] FIG. **8**B shows the policy table corresponding to a case where the first endurance type ET**1** is a time and the second endurance type ET**2** is a read count RC.

[0132] When the risk level is the first level, that is, when the endurance manager determines that the difference value does not exceed a second threshold value Th**2**, the endurance manager may reset read levels of the memory cells of the non-volatile memory device. For example, the endurance manager may load reset information stored in the non-volatile memory device and may reset the read levels of the memory cells. The reset information may include an initial value of each of the read levels of the memory cells. The reset information may be defined in advance and may be stored in the non-volatile memory device.

[0133] When the risk level is the second level, that is, when the endurance manager determines that the difference value exceeds the second threshold value Th**2**, the endurance manager may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0134] FIG. **8**C shows the policy table corresponding to a case where the first endurance type ET**1** is the erase count EC or the read count RC and the second endurance type ET**2** is the number of times of reclaim.

[0135] When the risk level is the first level, that is, when the endurance manager determines that the difference value does not exceed a third threshold value Th**3**, the endurance manager may control the non-volatile memory device such that only a restricted operation is performed. For example, the endurance manager may allow the non-volatile memory device to perform only the read operation.

[0136] When the risk level is the second level, that is, when the endurance manager determines that the difference value exceeds the third threshold value Th**3**, the endurance manager may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0137] FIG. **8**D shows the policy table corresponding to a case where the first endurance type ET**1** is the erase count EC or the read count RC and the second endurance type ET**2** is the program fail count or the erase fail count.

[0138] When the risk level is the first level, that is, when the endurance manager determines that the difference value does not exceed a fourth threshold value Th**4**, the endurance manager may control the non-volatile memory device such that only a restricted operation is performed. For example, the endurance manager may allow the non-volatile memory device to perform only the read operation.

[0139] When the risk level is the second level, that is, when the endurance manager determines that the difference value exceeds the fourth threshold value Th**4**, the endurance manager may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0140] FIG. **8**E shows the policy table corresponding to a case where the first endurance type ET**1** is the erase count EC or the read count RC and the second endurance type ET**2** is the number of times of execution of an error correction code or the error correction step.

[0141] When the risk level is the first level, that is, when the endurance manager determines that the difference value does not exceed a fifth threshold value Th**5**, the endurance manager may control the non-volatile memory device such that only a restricted operation is performed. For example, the endurance manager may allow the non-volatile memory device to perform only the read operation.

[0142] When the risk level is the second level, that is, when the endurance manager determines that the difference value exceeds the fifth threshold value Th**5**, the endurance manager may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0143] FIG. **8**F shows the policy table corresponding to a case where the first endurance type ET**1** is the erase count EC or the read count RC and the second endurance type ET**2** is the number of bad blocks.

[0144] When the risk level is the first level, that is, when the endurance manager determines that the difference value does not exceed a sixth threshold value Th**6**, the endurance manager may control the non-volatile memory device such that only a restricted operation is performed. For example, the endurance manager may allow the non-volatile memory device to perform only the read operation.

[0145] When the risk level is the second level, that is, when the endurance manager determines that the difference value exceeds the sixth threshold value Th**6**, the endurance manager may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0146] For convenience of description, the description is given using an example in which the endurance manager compares the difference value with one threshold value and performs one of two recovery operations depending on a comparison result, but the present disclosure is not limited thereto. The endurance manager may compare the difference value with two or more threshold values and may perform one of three or more recovery operations. Also, a kind of each of the recovery operations is not limited thereto. For example, the endurance manager may control the non-volatile memory device such that only a restricted operation is performed and may notify the host that the non-volatile memory device is controlled to perform only the restricted operation. As another example, the endurance manager may report a situation to the host for the purpose of debugging in a development process of a product which includes the storage device according to the present disclosure.

[0147] Also, the recovery action table is described as including the policy tables of FIGS. **8**A to **8**F, but the present disclosure is not limited thereto. The recovery action table may correspond to a table implemented by merging the policy tables of FIGS. **8**A to **8**F for each column. In this case, the recovery manager may determine the recovery operation by referring to the recovery action table based on the first endurance type, the second endurance type, and the risk level.

[0148] FIG. **9** is a flowchart describing an operating method of a storage device according to an embodiment of the present disclosure. Referring to FIGS. **1** and **9**, a storage device may include the storage controller **110** and the non-volatile memory device **120**.

[0149] In operation S**110**, the non-volatile memory device **120** may provide a response to the storage controller **110**. The response may correspond to at least one memory operation performed by the non-volatile memory device **120**.

[0150] In operation S**120**, the storage controller **110** may generate the first value of the first endurance type ET**1** and the second value of the second endurance type ET**2** based on the response.

[0151] In some embodiments, operation S**120** may include obtaining a first existing value of a first endurance type and a second existing value of a second endurance type by referring to an endurance data table, generating a first value by adding a value (e.g., the number of times) of the first endurance type included in a response to the first existing value, and generating a second value by adding a value (e.g., the number of times) of the second endurance type included in the response to the second existing value. The number of times of the first endurance type and the number of times of the second endurance type included in the response may be an integer which is not negative.

[0152] In some embodiments, the storage controller **110** may update the endurance data table based on the first value and the second value. The first existing value of the endurance data table may be changed to the first value, and the second existing value thereof may be changed to the second value.

[0153] In operation S**130**, the storage controller **110** may obtain a third value by applying the first value of the first endurance type ET**1** to the threshold function.

[0154] In some embodiments, the threshold function may be defined based on a correlation between the first endurance type ET**1** and the second endurance type ET**2**. The threshold function may be defined in advance.

[0155] In operation S**140**, the storage controller **110** may determine that the second value of the second endurance type ET**2** is greater than or equal to the third value.

[0156] In operation S**150**, the storage controller **110** may provide a recovery command to the non-volatile memory device **120**. The recovery command may indicate all commands provided to the non-volatile memory device **120** while the storage controller **110** performs the recovery operation.

[0157] In some embodiments, operation S**150** may include determining the recovery operation corresponding to the first endurance type, the second endurance type, and a difference value between the second value and the third value, based on determining that the second value is greater than or equal to the third value, and providing the recovery command corresponding to the determined recovery operation to the non-volatile memory device **120**.

[0158] In some embodiments, operation S**150** may include determining whether the difference value between the second value and the third value exceeds the threshold value determined based on a combination of the first endurance type and the second endurance type, performing the strong recovery action as the recovery operation when the difference value exceeds the threshold value, and performing the weak recovery action as the recovery operation when the difference value does not exceed the threshold value.

[0159] In some embodiments, operation S**150** may include calculating the difference value between the second value and the third value, determining a target risk level among a plurality of risk levels depending on whether the difference value exceeds the threshold value, determining a target recovery action among a plurality of recovery actions by referring to a recovery action table based on the target risk level, and providing a command corresponding to the target recovery action to the non-volatile memory device **120**.

[0160] For example, the recovery command may include one or more read commands. As another example, the recovery command may be a command for stopping an operation of the non-volatile memory device **120**.

[0161] FIG. **10** is a flowchart describing an operating method of a storage controller according to one or more example embodiments of present disclosure. An operating method of the storage controller **110** of FIG. **1** will be described with reference to FIG. **10**.

[0162] In operation S**210**, a storage controller may provide a command indicating a memory operation to a non-volatile memory device. For example, the storage controller may provide the non-volatile memory device with one of a read command indicating a read operation and a write command indicating a write operation.

[0163] In operation S**220**, the storage controller may receive a response corresponding to the memory operation from the non-volatile memory device.

[0164] In operation S**230**, the storage controller may generate a first value of the first endurance type ET**1** and a second value of the second endurance type ET**2** based on the response.

[0165] In operation S**240**, the storage controller may obtain a third value by applying the first value of the first endurance type ET**1** to the threshold function.

[0166] In operation S**250**, the storage controller may determine whether the second value of the second endurance type ET**2** is greater than or equal to the third value. When it is determined that the second value is smaller than the third value, the storage controller may perform operation S**210**. In other words, the storage controller may determine that the non-volatile memory device is in a normal state and may control the non-volatile memory device such that the memory operation is continuously performed. When it is determined that the second value is greater than or equal to the third value, the storage controller may perform operation S**260**.

[0167] In operation S**260**, the storage controller may determine whether a difference value between the second value and the third value exceeds a threshold value.

[0168] In some embodiments, the storage controller may determine the threshold value based on the first endurance type ET**1** and the second endurance type ET**2**. For example, the storage controller may store at least one combination of two different endurance types belonging to each of endurance type groups and threshold values each corresponding to one of the at least one combination.

[0169] The storage controller may perform operation S**262** based on determining that the difference value exceeds the threshold value. The storage controller may perform operation S**261** based on determining that the difference value not exceed the threshold value.

[0170] In operation S**261**, the storage controller may perform the weak recovery action as the recovery operation. For example, the storage controller may allow the non-volatile memory device to perform only the read operation.

[0171] In operation S**262**, the storage controller may perform the strong recovery action as the recovery operation. For example, the storage controller may stop an operation of the non-volatile memory device and may notify the host that the operation is stopped.

[0172] FIG. **11** is a diagram describing an operating method of a storage controller to monitor three or more endurance types in detail, according to one or more example embodiments of the present disclosure. Referring to FIG. **11**, a storage controller may monitor three or more endurance types and may use two or more threshold functions.

[0173] The host **11**, a storage controller **210**, and a non-volatile memory device **220** of FIG. **11** may respectively correspond to the host **11**, the storage controller **110**, and the non-volatile memory device **120** of FIG. **1**. An endurance manager **211**, a monitoring manager **211***a*, a recovery manager **211***b*, an endurance data table **212**, and a recovery action table **213** of FIG. **11** may respectively correspond to the endurance manager **111**, the monitoring manager **111***a*, the recovery manager **111***b*, the endurance data table **112**, and the recovery action table **113** of FIG. **5**.

[0174] The monitoring manager **211***a* may generate endurance data, based on the response RP received from the non-volatile memory device **220**. The endurance data may include a plurality of endurance types and values respectively corresponding to the plurality of endurance types.

[0175] For example, the endurance data may include the first endurance type ET**1**, the second endurance type ET**2**, and a third endurance type ET**3** and may include the first value x_k of the first endurance type ET**1**, the second value y_k of the second endurance type ET**2**, and a third value z_k of the third endurance type ET**3**.

[0176] The monitoring manager **211***a* may obtain existing endurance data from the endurance data table **212** and may generate the first to third values x_k, y_k, and z_k by updating the existing endurance data based on the response RP. The monitoring manager **211***a* may update the endurance data table **212** based on the first to third values x_k, y_k, and z_k.

[0177] The monitoring manager **211***a* may store a plurality of threshold functions. For example, the monitoring manager **211***a* may store a first threshold function y(x) and a second threshold function z(x). The first threshold function y(x) may be defined based on a correlation between the first endurance type ET**1** and the second endurance type ET**2**. The second threshold function z(x) may be defined based on a correlation between the first endurance type ET**1** and the third endurance type ET**3**. Although not illustrated, the monitoring manager **211***a* may further store a third threshold function which is defined based on a correlation between the second endurance type ET**2** and the third endurance type ET**3**.

[0178] The monitoring manager **211***a* may obtain a fourth value $y(x_k)$ by applying the first value x_k to the first threshold function y(x). The monitoring manager **211***a* may determine whether the second value y_k is greater than or equal to the fourth value $y(x_k)$.

[0179] Also, the monitoring manager **211***a* may obtain a fifth value $z(x_k)$ by applying the first value x_k to the second threshold function z(x). The monitoring manager **211***a* may determine whether the third value z_k is greater than or equal to the fifth value $z(x_k)$.

[0180] The monitoring manager **211***a* may generate the recovery request signal RRS, based on determining that the second value y_k is greater than or equal to the fourth value $y(x_k)$ and/or determining that the third value z_k is greater than or equal to the fifth value $z(x_k)$.

[0181] When the second value y_k is greater than or equal to the fourth value $y(x_k)$, the recovery request signal RRS may include the first endurance type ET**1**, the second endurance type ET**2**, and a first difference value. The first difference value may indicate a difference between the second value y_k and the fourth value $y(x_k)$.

[0182] When the third value z_k is greater than or equal to the fifth value $z(x_k)$, the recovery request signal RRS may include the first endurance type ET**1**, the third endurance type ET**3**, and a second difference value. The second difference value may indicate a difference between the third value z_k and the fifth value $z(x_k)$.

[0183] When the second value y_k is greater than or equal to the fourth value $y(x_k)$ and when the third value z_k is greater than or equal to the fifth value $z(x_k)$, the recovery request signal RRS may include the first endurance type ET**1**, the second endurance type ET**2**, the third endurance type ET**3**, and the first difference value, and the second difference value.

[0184] The recovery manager **211***b* may determine a recovery operation based on the recovery request signal RRS and may perform the determined recovery operation. For convenience of description, a case where endurance data are out of both the tendency of the first threshold function and the tendency of the second threshold function will be described below. When the endurance data are out of only one of the two tendencies, the recovery manager **211***b* operates to be similar to the recovery manager **111***b* of FIG. **5**.

[0185] The recovery manager **211***b* may determine a first risk level, based on the first endurance type ET**1**, the second endurance type ET**2**, and the first difference value of the recovery request signal RRS.

[0186] In some embodiments, the recovery manager **211***b* may store at least one combination of two different endurance types belonging to each endurance type group and a plurality of threshold values each corresponding to one of the at least one combination. The recovery manager **211***b* may determine a first threshold value corresponding to a combination of the first endurance type ET**1** and the second endurance type ET**2**. The recovery manager **211***b* may compare the first difference value and the first threshold value and may determine the first risk level based on a result of comparison.

[0187] For example, when the first difference value does not exceed the first threshold value, the recovery manager **211***b* may determine the first risk level to be a first level. In contrast, when the first difference value exceeds the first threshold value, the recovery manager **211***b* may determine the first risk level to be a second level.

[0188] As in the above description, the recovery manager **211***b* may determine a second risk level, based on the first endurance type ET**1**, the third endurance type ET**3**, and the second difference value of the recovery request signal RRS.

[0189] The recovery manager **211***b* may determine a first recovery action corresponding to the first risk level by referring to a first policy table corresponding to the first endurance type ET**1** and the second endurance type ET**2** from among a plurality of policy tables of the recovery action table **213**. The plurality of policy tables correspond to the policy table of FIGS. **5** and **8**.

[0190] Likewise, the recovery manager **211***b* may determine a second recovery action corresponding to the second risk level by referring to a second policy table corresponding to the first endurance type ET**1** and the third endurance type ET**3** from among the plurality of policy tables.

[0191] The recovery manager **211***b* may perform the first recovery action and the second recovery action as the recovery operation. When the first recovery action is the same as the second recovery action, the recovery manager **211***b* may perform the first recovery action once as the recovery operation.

[0192] According to one or more example embodiments of the present, an operating method of the storage controller **210** will be described in detail.

[0193] In a first operation ①, the monitoring manager **211***a* may receive the response RP from the non-volatile memory device **220**.

[0194] In a second operation ②, the monitoring manager **211***a* may generate the first value x_k of the first endurance type ET**1**, the second value y_k of the second endurance type ET**2**, and the third value z_k of the third endurance type ET**3**, based on the response RP.

[0195] In a third operation ③, the monitoring manager **211***a* may determine whether the recovery operation is necessary. When it is determined that the second value y_k is greater than or equal to the fourth value $f(x_k)$ obtained by applying the first value x_k to the first threshold function fx, the monitoring manager **211***a* may determine that the recovery operation is required. Additionally or alternatively, when it is determined that the third value z_k is greater than or equal to the fifth value $z(x_k)$ obtained by applying the first value x_k to the second threshold function z(x), the monitoring manager **211***a* may determine that the recovery operation is required.

[0196] In a fourth operation ④, the monitoring manager **211***a* may provide the recovery request signal RRS to the recovery manager **211***b*, based on determining that the recovery operation is required.

[0197] In a fifth operation ⑤, the recovery manager **211***b* may perform the recovery operation. The recovery manager **211***b* may determine the recovery operation by referring to the recovery action table **213** based on the recovery request signal RRS.

[0198] According to an embodiment of the present disclosure, a storage device performing a recovery operation by using endurance data and an operation method thereof are provided.

[0199] Also, the damage to endurance of the non-volatile memory device may be prevented preemptively and efficiently by monitoring variables affecting the endurance and/or affected by the endurance, determining whether to perform a recovery operation in consideration of a correlation between the variables, and differently performing the recovery operation depending on a degree to which the monitored variables are out of the correlation. Accordingly, a storage device whose endurance is improved may be provided.

[0200] At least one of the components, elements, modules and units (collectively "components" in this paragraph) represented by a block or an equivalent indication in the drawings including FIG. 1 described above may use a direct circuit structure, such as a memory, a processor, a logic circuit, a look-up table, etc. that may execute the respective functions through controls of one or more microprocessors or other control apparatuses. Alternatively or additionally, at least one of these components may be specifically embodied by a module, a program, or a part of code, which is stored in an internal memory of the storage controller **110** or an external memory, and contains one or more executable instructions for performing the above-described functions, and executed by one or more microprocessors or other controller included in the storage controller **110**. Further, at least one of these components may include or may be implemented by a processor such as a central processing unit (CPU), graphic processing unit (GPU), another type of microprocessor, or the like in the storage controller **110** that performs the above-described functions. Two or more of these components may be combined into one single component which performs all operations or functions of the combined two or more components. Also, at least part of functions of at least one of these components may be performed by another of these components. Functional aspects of the above example embodiments may be implemented in algorithms that execute on one or more processors.

[0201] While the present disclosure has been described with reference to example embodiments thereof, it will be apparent to those of ordinary skill in the art that various changes and modifications may be made thereto without departing from the spirit and scope of the present disclosure as set forth in the following claims and equivalents.

What is claimed is:

1. An operating method of a storage controller configured to communicate with a non-volatile memory device, the method comprising:

receiving a memory response from a non-volatile memory device;

generating a first value of a first endurance type and a second value of a second endurance type, based on the memory response, the first endurance type and the second endurance type being related to an endurance of the non-volatile memory device;

obtaining a third value by applying the first value to a threshold function, wherein the threshold function is defined based on a correlation between the first endurance type and the second endurance type;

determining whether the second value is greater than or equal to the third value; and

performing a recovery operation in response to determining that the second value is greater than or equal to the third value.

2. The method of claim **1**, wherein the performing the recovery operation includes:

determining the recovery operation, among a plurality of recovery operations, corresponding to the first endurance type, the second endurance type, and a difference value between the second value and the third value, in response to determining that the second value is greater than or equal to the third value.

3. The method of claim **2**, wherein the storage controller is configured to communicate with a host,

wherein the determining the recovery operation among the plurality of recovery operations includes:

determining whether the difference value exceeds a threshold value, wherein the threshold value is determined based on a combination of the first endurance type and the second endurance type; and

determining a first recovery action as the recovery operation based on the difference value exceeding the threshold value and determining a second recovery action as the recovery operation based on the difference value not exceeding the threshold value.

4. The method of claim **3**, wherein the first recovery action indicates stopping an operation of the non-volatile memory device and notifying the host that the operation of the non-volatile memory device is stopped.

5. The method of claim **3**, wherein the second recovery action indicates controlling the non-volatile memory device to perform a restricted operation.

6. The method of claim **5**, wherein the restricted operation indicates a read operation.

7. The method of claim **5**, wherein the first endurance type corresponds to a time,

wherein the second endurance type corresponds to a read count, and

wherein the restricted operation indicates an operation of resetting read levels of memory cells of the non-volatile memory device.

8. The method of claim **2**, wherein the storage controller includes a recovery action table,

wherein the recovery action table includes a plurality of risk levels corresponding to a combination of the first endurance type and the second endurance type, and a plurality of recovery actions respectively corresponding to the plurality of risk levels,

the determining the recovery operation among the plurality of recovery operations includes:

obtaining the difference value between the second value and the third value;

determining a target risk level among the plurality of risk levels, based on whether the difference value exceeds a threshold value, wherein the threshold value is determined based on the combination of the first endurance type and the second endurance type; and

determining a target recovery action corresponding to the target risk level from among the plurality of recovery actions as the recovery operation, by referring to the recovery action table.

9. The method of claim **1**, wherein each of the first endurance type and the second endurance type indicates an event that affects the endurance of the non-volatile memory device and/or an event affected by the endurance,

wherein the memory response includes a value of "n" of the first endurance type and a value of "m" of the second endurance type, and

wherein each of n and m is 0 or a positive integer.

10. The method of claim **9**, wherein the first endurance type and the second endurance type indicate two different ones among an erase count, a read count, a program count, a time, a number of times of reclaim, a number of times of execution of an error correction code, an error correction success step, a number of bad blocks, a program fail count, and an erase fail count.

11. The method of claim **9**, wherein the storage controller includes an endurance data table, and

wherein the endurance data table includes a first existing value of the first endurance type and a second existing value of the second endurance type.

12. The method of claim **11**, wherein the generating the first value and the second value includes:

obtaining the first existing value of the first endurance type and the second existing value of the second endurance type by referring to the endurance data table;

generating the first value of the first endurance type by adding the value of "n" to the first existing value; and

generating the second value of the second endurance type by adding the value of "m" to the second existing value.

13. The method of claim **12**, wherein the generating the first value and the second value further includes:

updating the endurance data table, based on the first value of the first endurance type and the second value of the second endurance type.

14. The method of claim **1**, wherein the first endurance type represents a time, and

wherein the second endurance type indicates one of an erase count and a read count.

15. The method of claim **1**, wherein the first endurance type indicates one of an erase count and a read count, and

wherein the second endurance type indicates one of a program count, a number of times of reclaim, a number of times of execution of an error correction code, an error correction success step, a number of bad blocks, a program fail count, and an erase fail count.

16. An operating method of a storage controller configured to communicate with a non-volatile memory device, the method comprising:

receiving a memory response from a non-volatile memory device;

generating a first value of a first endurance type, a second value of a second endurance type, and a third value of a third endurance type, based on the memory response, the first endurance type, the second endurance type, and the third endurance type being related to an endurance of the non-volatile memory device;

obtaining a fourth value by applying the first value to a first threshold function, wherein the first threshold function is defined based on a correlation between the first endurance type and the second endurance type;

obtaining a fifth value by applying the first value to a second threshold function, wherein the second threshold function is defined based on a correlation between the first endurance type and the third endurance type; and

performing a recovery operation, based on at least one of determining that the second value is greater than or equal to the fourth value or determining that the third value is greater than or equal to the fifth value.

17. The method of claim **16**, wherein the performing the recovery operation includes:

determining a first recovery action corresponding to the first endurance type, the second endurance type, and a first difference value between the second value and the fourth value from among a plurality of recovery actions in response to determining that the second value is greater than or equal to the fourth value;

determining a second recovery action corresponding to the first endurance type, the third endurance type, and a second difference value between the third value and the fifth value from among the plurality of recovery actions in response to determining that the third value is greater than or equal to the fifth value; and

performing at least one of the first recovery action or the second recovery action as the recovery operation.

18. The method of claim **17**, wherein the determining the first recovery action includes:

determining whether the first difference value exceeds a first threshold value, wherein the first threshold value is determined based on a combination of the first endurance type and the second endurance type; and

determining a first recovery action as the first recovery action based on the first difference value exceeding the first threshold value and determining a second recovery action as the first recovery action based on the first difference value not exceeding the first threshold value, and

wherein the determining the second recovery action includes:

determining whether the second difference value exceeds a second threshold value, wherein the second threshold value is determined based on a combination of the first endurance type and the third endurance type; and

determining a first recovery action as the second recovery action based on the second difference value exceeding the second threshold value and determining a second recovery action as the second recovery action based on the second difference value not exceeding the second threshold value.

19. The method of claim **16**, wherein the first endurance type, the second endurance type, and the third endurance type respectively indicate three different ones among an erase count, a read count, a program count, a time, a number of times of reclaim, a number of times of execution of an error correction code, an error correction success step, a number of bad blocks, a program fail count, and an erase fail count.

20. A storage device comprising:

a non-volatile memory device; and

a storage controller,

wherein the storage controller includes:

a monitoring manager configured to receive a memory response from the non-volatile memory device, to generate a first value of a first endurance type and a second value of a second endurance type based on the memory response, to obtain a third value by applying the first value to a threshold function, and to generate a recovery request signal in response to determining that the second value is greater than or equal to the third value; and

a recovery manager configured to perform a recovery operation in response to the recovery request signal,

wherein the first endurance type and the second endurance type are related to an endurance of the non-volatile memory device, and

wherein the threshold function is defined based on a correlation between the first endurance type and the second endurance type.

* * * * *